ASWATHY S.

Robotics Software Engineer

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PROFESSIONAL SUMMARY

Robotics Software Engineer with an M.Tech in Robotics and Automation (CGPA 9.2) and hands-on experience developing autonomous mobile robots from simulation to deployment. Proficient in Python, ROS/ROS 2, with demonstrated expertise in SLAM, navigation stack optimization, and path planning algorithms. Experienced in end-to-end robot development through DRDO-sponsored defense projects, robust error handling, and sensor fusion for real-world autonomous systems. Demonstrated expertise in implementing and tuning Extended Kalman Filters (EKF) for multi-sensor integration (LiDAR, IMU, Odometry) to achieve robust localization in complex environments. Passionate about advancing robust localization and perception technologies for off-road and industrial autonomous platforms.

CORE COMPETENCIES

- **Navigation & Planning:** SLAM (Gmapping, Cartographer), AMCL Localization, Global/Local Planners, Coverage Path Planning.
- **Software Development:** Python, ROS/ROS 2, Linux/Ubuntu, Git, Docker, Debugging Real-Time Systems.
- **System Integration:** Sensor Fusion (LiDAR, IMU), Simulation-to-Hardware Pipelines.
- **Perception & AI:** Computer Vision (OpenCV), Deep Learning (PyTorch).

RESEARCH & PROFESSIONAL EXPERIENCE

Project Associate (DRDO Sponsored Project — CARS 95-DRDO, CAIR) *Oct 2025 – Sep 2026*
Indian Institute of Technology (IIT), Madras

- Designed and implemented **simultaneous exploration and coverage path planning** algorithms for autonomous robots in unknown environments, optimizing for real-time performance and robustness.
- Built the full autonomy stack using **ROS 2**—including SLAM, and custom planners—achieving reliable navigation under varying conditions.
- Developed high-fidelity **Gazebo simulation** frameworks to pre-validate algorithms.
- Collaborated with hardware and controls teams to deploy software on robot platforms; conducted systematic **debugging** of runtime issues (localization drift, path oscillations, sensor noise).

EDUCATION

M.Tech in Robotics and Automation *2024 – 2026*
College of Engineering, Trivandrum (CET) (APJ Abdul Kalam Technological University) **CGPA: 9.2**

- *Relevant Coursework:* Kinematics & Dynamics, Advanced Control Systems, Machine Learning, Deep Learning, Embedded Systems.

B.Tech in Mechanical Engineering *2020 – 2024*
College of Engineering, Perumon (APJ Abdul Kalam Technological University) **CGPA: 7.42**

- *Relevant Coursework:* Mechatronics, Control Systems, Kinematics, CAD/CAM, Fluid Mechanics.

TECHNICAL SKILLS

- **Languages:** Python.

- **Robotics Frameworks:** ROS 1/2, Gazebo, Rviz, Navigation Stack.
- **Mapping & Localization:** SLAM, AMCL, Odometry Calibration.
- **Planning & Control:** A*, Dijkstra, TEB, DWB, Coverage Path Planning, PID, Motion Control.
- **Sensors & Peripherals:** 2D LiDAR, IMU, Arduino, Raspberry Pi.
- **Tools:** Linux (Ubuntu), Git, OpenCV, PyTorch.

KEY PROJECTS

Autonomous Indoor Navigation Robot (AGV)

ROS1, SLAM, Navigation Stack

- Architected and developed a fully autonomous indoor delivery robot using ROS 1 , implementing the complete autonomy stack from perception to motion control.
- Built SLAM-based mapping (Gmapping) and AMCL localization pipeline, achieving robust pose estimation through multi-sensor fusion (LiDAR + IMU).
- Configured and optimized the ROS Navigation Stack—tuned global (A*) and local (TEB) planners for smooth path tracking, dynamic obstacle avoidance, and precise goal reaching.
- Validated system in both Gazebo simulation and hardware, conducting systematic debugging of localization drift, path oscillations, and sensor noise issues.

Deep Learning–Based Pothole Detection

Python, PyTorch, OpenCV

- Engineered a CNN using PyTorch for real-time road anomaly detection .
- Developed data preprocessing pipelines with image augmentation and optimized model inference for edge deployment.
- Achieved high detection accuracy; documented software architecture and testing procedures for reproducibility

Autonomous Line Follower with PID Control

Embedded C, Control Theory

- Implemented and tuned PID controllers on an IR array–based platform, demonstrating hands-on control systems intuition applicable to mobile robot motion control.

ACHIEVEMENTS

- Secured Third Place in an Intercollegiate Handball Tournament (Demonstrates team spirit and leadership).
- Coordinated arts and cultural events at the college level (Demonstrates organizational skills).